Earners AI and ML by Aditya Jain sir - Lecture 11

Link to Telegram Community: https://t.me/EarnersAdityaJain

Recap - List

```
a = [4,6,7,2,1]
```

```
type(a)
```

```
list
```

```
n = len(a)
print(n)
```

```
5
```

```
for i in range(n):
  print(a[i])
```

```
4
6
7
2
1
```

```
for elem in a:
  print(elem)
```

```
4
6
7
2
1
```

```
#Print the elements of the list in reverse order

for i in range(n):
  print(a[n-1-i])
```

```
1
2
7
6
4
```

```
numbers = [1, 2, 3, 4]
n = len(numbers)

for i in range(n):
    numbers[i] = numbers[i] + i

print(numbers)
```

```
[1, 3, 5, 7]
```

```
marks = [65, 75, 85]

total = 0

for m in marks:
    if m >= 70:
        total += m

print(total)
```

```
160
```

```
nums = [2, 4, 6]

nums[0] = nums[2]

nums[2] = nums[1]

print(nums)
```

```
[6, 4, 4]
```

```
names = ["Amit", "Neha", "Aditya"]

longest = names[0]

for name in names:
    if len(name) > len(longest):
        longest = name

print(longest)
```

```
Aditya
```

```
data = [10, 20, 30]

for i in range(len(data)):
    data[i] = data[i] + 10

print(max(data))

print(min(data))
```

```
40
20
```

```
numbers = [5, 10, 15, 20]

total = 0

for i in range(len(numbers)):
    if i % 2 == 0:
        total += numbers[i]

print(total)
```

```
20
```

List Slicing

```
a = [5, 7, 4, 8, 2, 1, 9]
print(a)
```

```
[5, 7, 4, 8, 2, 1, 9]
```

```
print(a[1:4])
```

```
[7, 4, 8]
```

```
print(a[2:5])
```

```
[4, 8, 2]
```

```
print(a[:4])
```

```
[5, 7, 4, 8]
```

```
print(a[0:4])
```

```
[5, 7, 4, 8]
```

```
print(a[1:])
```

```
[7, 4, 8, 2, 1, 9]
```

```
print(a[ : ])
```

```
[5, 7, 4, 8, 2, 1, 9]
```

```
a
```

```
[5, 7, 4, 8, 2, 1, 9]
```

```
print(a[1:5:2])
```

```
[7, 8]
```

```
print(a[ : : 2])
```

```
[5, 4, 2, 9]
```

Strings

```
name = "AJ"
```

```
type(name)
```

```
str
```

```
a = "24"
type(a)
```

```
str
```

```
email = "aditya@aj.com"
```

```
#empty string:

s = ""
print(s)
```

```

```

```
len(s)
```

```
0
```

```
len("Aditya")
```

```
6
```

```
name = "Aditya"
print(name[0])
```

```
A
```

```
print(name[1])
```

```
d
```

```
print(name[-1])
```

```
a
```

```
#Print the given string in reverse order.
name = "Aditya"

n = len(name)
```

```
for i in range(n):
  print(name[n-1-i], end ="")
```

```
aytidA
```

```
print(name[1:4])
```

```
dit
```

```
# print(name[::-1]) --> special case
```

```
aytidA
```

```
print(name[::])
```

```
Aditya
```

```
for i in range(n):
  print(name[i])
```

```
A
d
i
t
y
a
```

```
for elem in name:
  print(elem)
```

```
A
d
i
t
y
a
```

```
for c in name:
  print(c)
```

```
A
d
i
t
y
a
```

```
4 + 6
```

```
10
```

```
"4"+ "6"
```

```
'46'
```

```
"Aditya" + " Jain"
```

```
'Aditya Jain'
```

Quiz

```
word = "Python"

print(word[3])
```

```
h
```

```
text = "Machine"

print(text[-2])
```

```
n
```

```
name = "Rahul"

print(len(name))
```

```
5
```

```
first = "Data"

second = "Science"

print(first + second)
```

```
DataScience
```

```
print("AI" * 3)
```

```
AIAIAI
```

```
"aj"*3
```

```
'ajajaj'
```

```
print("Aditya")
print("-"*20)
print("Jain")
```

```
Aditya
--------------------
Jain
```

```
word = "Python"

print("P" in word)
```

```
True
```

```
word = "Python"

print("p" in word)a
```

```
False
```

```
word = "Python"

print("z" not in word)
```

```
True
```

```
word = "Python"

print(word[0] + word[-1])
```

```
Pn
```

```
word = "Python"

word[0] = "J"

print(word)
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
/tmp/ipykernel_1833/2456199997.py in <cell line: 0>()
      1 word = "Python"
      2
----> 3 word[0] = "J"
      4
      5 print(word)

TypeError: 'str' object does not support item assignment
```

Next steps: Explain error

```
#Strings are immutable --> cant be modified.
```

```
a = [4,7,8]
a[1] = 10
print(a)
```

```
[4, 10, 8]
```

```
word = "Code"

for i in range(len(word)):
    print(i, word[i])
```

```
0 C
1 o
2 d
3 e
```

```
text = "Python"

i = 0

while i < len(text):
    print(text[i])
    i += 2
```

```
P
t
o
```

```
word = "Python"

count = 0

for ch in word:
    if ch in "aeiouAEIOU":
        count += 1

print(count)
```

```
1
```

```
text = "Programming"

count = 0

for ch in text:
    if ch == "m":
        count += 1

print(count)
```

```
2
```

```
word = "Python"

result = ""

for ch in word:
    result = ch + result
    print(result)
```

```
P
yP
tyP
htyP
ohtyP
nohtyP
```

```
text = "Data Science "

spaces = 0

for ch in text:
    if ch == " ":
        spaces += 1

print(spaces)
```

```
2
```

```
word = "Python"

total = 0

for ch in word:
    total += 1

print(total == len(word))
```

```
True
```

HW - DPP- Tricky Advanced Questions

```
text = "Programming"

count = 0

for i in range(len(text)):
    if text[i] == text[-1]:
        count += 1

print(count)
```

```
word = "PYTHON"

result = ""

for i in range(len(word)):
    if i % 2 == 1:
        result += word[i]

print(result)
```

```
text = "Data"

result = ""

for ch in text:
    result = ch + result

print(result)
```

```
word = "Machine"

count = 0

for ch in word:
    if ch not in "aeiouAEIOU":
        count += 1

print(count)
```

```
word = "Python"

i = len(word) - 1

while i >= 0:
    print(word[i], end="")
    i -= 2
```

```
text = "AI"

result = ""

for i in range(4):
    result = result + text

print(len(result))
```

```
word = "Computer"

count = 0

for ch in word:
    if ch < "m":
        count += 1

print(count)
```

```
text = "Python"

result = ""

for i in range(len(text)):
    if text[i] in "aeiouAEIOU":
        result += text[i]

print(result)
```

```
word = "LEVEL"

same = 0

for i in range(len(word)):
    if word[i] == word[-(i + 1)]:
        same += 1

print(same)
```

```
text = "Python"

result = ""

for i in range(len(text)):
    if i < len(text) // 2:
        result += text[i]

print(result)
```